

# AI Visualization Tool — Error Handling & Resilient UX Framework

**Company:** Reform AI

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**Tool URL:** <https://app.reform-ai.com/es/home/all-listings?lang=en>

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## 1. Objective

The objective of this document is to define a **robust error handling and fallback UX framework** for the AI Visualization Tool, ensuring:

- Continuity of the user journey
- Elimination of dead ends in the buyer flow
- Increased reliability perception of the product
- Higher conversion and engagement rates

This framework focuses on maintaining **forward user momentum**, even when failures occur during AI generation or input processing.

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## 2. Scope of the Flow

The evaluated flow includes the full AI visualization journey:

1. User enters the platform
2. Navigates to **Reform-AI** section
3. Uploads base image
4. Selects:
  - Room type (mandatory — single selection)
  - Design style (mandatory — single selection)
5. Optionally adds:
  - Prompt / indication

- Moodboard (currently limited to 1 image)
  - Furniture (1 image)
- 6. Generates AI visualization
- 7. Interacts with results:
  - Regenerate design
  - Try different style
  - Download result
  - Expand image

This document addresses all **failure points within this flow**.

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## 3. UX Principles for Error Handling

The system must adhere to the following principles:

### 3.1 No Dead Ends

Users must never reach a state where:

- No action is available
  - No guidance is provided
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### 3.2 Actionable Feedback

All errors must:

- Clearly explain the issue
  - Provide a next step
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### 3.3 Silent Recovery When Possible

The system should:

- Automatically correct issues when feasible
  - Avoid interrupting the user unnecessarily
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### 3.4 Maintain User Trust

Errors must:

- Avoid technical jargon (e.g., “500 error”)
  - Communicate reliability and control
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### 3.5 Preserve User Inputs

User data (image, selections, prompt) must:

- Never be lost due to failure
  - Be reusable in retry scenarios
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## 4. Error Taxonomy

The following error categories have been identified based on current system behavior:

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### 4.1 Upload Errors

#### Description

Failures during:

- Base image upload
- Moodboard upload
- Furniture upload

#### Observed Issues

- Generic system failures (e.g., silent failure or unclear errors)
- No feedback on file compatibility

#### UX Requirements

- Display clear notification:
  - File format not supported

- File size too large
    - Upload failed
  - Provide immediate actions:
    - Retry upload
    - Replace file
    - Use optimized version
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## 4.2 Generation Failures

### Description

Failures during AI processing after clicking **Generate Design**

### Observed Issues

- No output generated
- System appears inactive or frozen
- Lack of feedback

### UX Requirements

- Display real-time status:
    - “Processing...”
    - “Still working...”
  - If failure occurs:
    - Notify user clearly
    - Offer retry option
    - Preserve all inputs
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## 4.3 Timeout / Slow Processing

### Description

Generation exceeds expected time or stalls without completion

### Observed Issues

- No timeout feedback
- User uncertainty about system state

## UX Requirements

- Define expected generation time range
  - If exceeded:
    - Display notification:
      - “This is taking longer than expected”
  - Provide options:
    - Continue waiting
    - Retry generation
    - Adjust inputs
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## 4.4 Non-Responsive Generation (No Output Produced)

### Description

System completes process but does not return a valid image

### UX Requirements

- Notify user:
    - “We couldn’t generate a result for this configuration”
  - Provide recovery options:
    - Regenerate with same inputs
    - Modify prompt
    - Change style
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## 4.5 Bad Outputs (Low Quality / Inconsistent Results)

### Description

Generated results that:

- Break spatial logic
- Distort furniture
- Alter background inconsistently

### UX Requirements

- Avoid labeling as “error”

- Present as alternative outcome
  - Provide options:
    - Regenerate
    - Try different style
    - Adjust prompt
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## 4.6 Moodboard Cropping / Aspect Ratio Issue

### Description

When moodboard image aspect ratio differs from base image:

- System automatically crops image
- Loss of composition and proportion

### Impact

- Reduced influence of moodboard
- Misalignment with user expectations

### UX Requirements

- Notify user:
    - “Moodboard image was adjusted to fit the scene”
  - Provide control (future improvement):
    - Fit vs crop option
    - Preview before generation
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## 5. Fallback Strategies

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### 5.1 Upload Fallback

- Auto-compress large images
- Normalize formats when possible
- Retry upload automatically once

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## 5.2 Generation Fallback

- Auto-retry generation once internally
  - Adjust internal parameters if failure detected
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## 5.3 Timeout Fallback

- Extend processing window once
  - Offer manual retry
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## 5.4 Output Fallback

- Provide regeneration options
  - Encourage variation exploration
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## 5.5 Input Compatibility Fallback

- Automatically adjust:
    - Resolution
    - Aspect ratio
  - Inform user when adjustments occur
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# 6. Notification System Requirements

All errors and fallback events must include:

## 6.1 Visibility

- Inline notification within workspace
- Non-blocking but clearly visible

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## 6.2 Clarity

- Human-readable explanation
- No technical codes

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## 6.3 Actionability

Each notification must include:

- Retry
- Edit inputs
- Continue

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## 6.4 Persistence

- Notifications remain visible until user action

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# 7. Critical UX Improvements

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## 7.1 Replace Silent Failures

- Eliminate all cases where:
  - No output
  - No feedback

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## 7.2 Improve Upload Feedback

- Add validation before upload
  - Provide immediate feedback
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## 7.3 Improve Generation Transparency

- Add clear loading states
  - Show progress indicators
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## 7.4 Preserve Workflow Continuity

- Ensure:
    - Inputs remain editable
    - Users can retry without restarting
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## 7.5 Improve Moodboard Handling

- Avoid destructive cropping without feedback
  - Provide preview or adjustment options
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# 8. Impact on Buyer Flow

Current risks:

- User confusion during failures
- Perceived unreliability
- Drop-off during generation stage

With improvements:

- Increased trust in AI system
  - Reduced abandonment
  - Higher likelihood of:
    - Iteration
    - Exploration
    - Conversion
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## 9. Final Assessment

The AI Visualization Tool currently demonstrates strong capabilities, but **error handling and fallback UX are critical gaps** in achieving:

- Production-level reliability
  - Seamless buyer experience
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## 10. Recommendations Summary

- Implement structured error taxonomy
  - Replace all generic/system errors with user-facing messages
  - Introduce fallback logic for all failure points
  - Ensure no dead ends in user flow
  - Improve transparency in processing states
  - Add control and feedback for input transformations
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## 11. Conclusion

A robust error handling and fallback UX system is essential to:

- Protect the integrity of the user experience
- Maintain momentum in the buyer journey
- Reinforce Reform AI's positioning as a reliable AI-driven product

This framework ensures that **every failure becomes a recoverable state**, rather than a point of abandonment.

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*End of Document — Error Handling & Resilient UX Framework*